

Vol 8 Chapter 2 — Newton's Laws of Motion

Keeper (author pass — deep math/physics revision)

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Chapter 2 — Newton's Laws of Motion

Level 1 — one sentence

Newton's three laws — inertia ($F = 0 \Rightarrow$ constant velocity), $F = ma$, and action-reaction — together with Galilean relativity form the foundation of classical mechanics, and they emerge in BST as the substrate's Scale-2 effective dynamics for slow massive bodies in flat-space limit of the substrate $SO_0(5, 2)$ geometry.

Level 2 — graduate-physicist precision

2.1 The three laws

First law (inertia): A body in uniform motion (or at rest) continues so unless acted on by an external force.

Second law: $\vec{F} = m\vec{a} = m d^2\vec{r}/dt^2 = d\vec{p}/dt$ (the third form is fundamental — force = rate of change of momentum).

Third law: For every action there is an equal and opposite reaction: $\vec{F}_{12} = -\vec{F}_{21}$.

2.2 Inertial frames and Galilean invariance

An inertial frame is one in which Newton's first law holds (a free body moves with constant velocity). Two inertial frames are related by Galilean transformation:

$$\vec{r}' = \vec{r} - \vec{v}_0 t, \quad t' = t$$

Newton's laws are invariant under Galilean transformations — physics is the same in all inertial frames.

Substrate reading: at Scale 2 with $v \ll c$, the substrate's Lorentz invariance (Vol 7 Ch 7) reduces to Galilean. The substrate doesn't have absolute rest — only relative motion between K-type configurations matters.

2.3 Conservation of momentum

From Newton's second + third laws: total momentum $\vec{P} = \sum_i \vec{p}_i$ of an isolated system is conserved.

For collisions: - Elastic: kinetic energy also conserved - Inelastic: kinetic energy lost (converted to heat, deformation) - Completely inelastic: particles stick together

Substrate reading: momentum conservation is the substrate's Noether current from spatial translation symmetry (T2474 per Friday May 22 EOD framework; Vol 0 Ch 8).

2.4 Force types

Common forces in classical mechanics: - Gravitational: $\vec{F} = -GMm\hat{r}/r^2$ - Spring: $\vec{F} = -k\vec{x}$ (Hooke) - Friction: $\vec{F} = -\mu N$ opposing motion - Drag: $\vec{F} \propto -v$ (Stokes) or $-v^2$ (Newton) - Centripetal: $F = mv^2/r$ for circular motion - Lorentz: $\vec{F} = q(\vec{E} + \vec{v} \times \vec{B})$ — bridges to EM (Vol 7)

2.5 Worked example: projectile motion

A baseball thrown at angle $\theta = 45^\circ$, initial speed $v_0 = 30$ m/s, neglecting air drag:

- Horizontal: $x(t) = v_0 \cos \theta \cdot t$
- Vertical: $y(t) = v_0 \sin \theta \cdot t - (1/2)gt^2$

Range $R = v_0^2 \sin(2\theta)/g = (30)^2 \cdot 1/9.8 \approx 92$ m.

Max height: $h = v_0^2 \sin^2 \theta / (2g) = 900 \cdot 0.5 / 19.6 \approx 23$ m.

Time of flight: $t = 2v_0 \sin \theta / g \approx 4.3$ s.

2.6 K-audit anchors

- T2474 (Friday May 22, 2026): substrate momentum conservation Noether
- Volume 7 Ch 7: Lorentz $\rightarrow$ Galilean limit at low velocity
- Volume 0 Ch 8: substrate Noether currents

Level 3 — 5th-grader accessibility

Newton's laws are the foundation of mechanics: 1. Inertia: things keep doing what they're doing unless something pushes 2. $F = ma$: push something, it accelerates; bigger mass needs more push 3. Action-reaction: push something, it pushes back equally hard

In any non-accelerating frame, these laws hold. Physics doesn't care if you're standing still or moving — only relative motion matters (Galilean relativity). Momentum is conserved when no outside forces act. In BST, all of this comes out of the substrate's Scale-2 limit when objects are slow (much less than c) and big (decohered).

What comes next

Chapter 3 develops Lagrangian mechanics — the variational reformulation.

Where to look this up

- Newton 1687, *Principia*
- Goldstein Ch 1; Marion and Thornton, *Classical Dynamics*
- BST: T2474, Vol 7 Ch 7